Fiscal Year:	FY 2022	**Task Last Updated:**	FY 09/01/2021
PI Name:	Yang, Jiseon Ph.D.		
Project Title:	Microbial Social Behavior and Heritable Genetic or Epigenetic Changes Affected by the Spaceflight Environment: Understanding the Evolution of Microbial Interactions During Spaceflight		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Translational (Countermeasure) Potential		
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Zip Code:	85287-5401	**Congressional District:**	9
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2018 Space Biology (ROSBio) NNH18ZTT001N-FG. App B: Flight and Ground Space Biology Research
Start Date:	11/01/2019	**End Date:**	10/31/2022
No. of Post Docs:	0	**No. of PhD Degrees:**	0
No. of PhD Candidates:	0	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	1
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA ARC
Contact Monitor:	Griko, Yuri	**Contact Phone:**	650-604-0519
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Flight Program:			
Flight Assignment:			
Key Personnel Changes/Previous PI:	September 2021 report: A lab technician (Sara Koroli, B.S.) worked from October 12, 2020 (Monday) to Friday, May 28, 2021 (Friday).		
COI Name (Institution):			
Grant/Contract No.:	80NSSC19K1597		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	Microbes naturally exist as mixed-species communities where they actively communicate and interact with neighboring cells. They adapt and evolve over time within the mixed-species populations. While many studies have shown that the spaceflight environment (and ground-based spaceflight analogs) impacts microbial biological processes, most of the previous experiments were based on single-species cultures. It is still unclear how the spaceflight environment influences microbial social behaviors and how their interaction in the mixed-species community contributes to microbial evolutionary processes in the spaceflight environment. Understanding the effects of the spaceflight environment on the changes in microbial social behavior and the associating evolutionary mechanisms will be necessary for spaceflight design, operations, and astronaut health. It is crucial in regards to preparation for long-duration life support systems during future exploration missions far from Earth. Microbes within the International Space Station (ISS) are continuously exposed and adapted to the unique spaceflight environment, which includes microgravity, low fluid shear, and enhanced radiation. A variety of microbial isolates have been recovered from the ISS, including the potable water system. The ISS Water Recovery System recycles urine, condensation, and wastewater. Astronauts rely on the recycled potable water for survival. Most of the microbes recovered from ISS potable water are known biofilm formers, which can increase risks for crew infections or damage to life support system function and integrity. Therefore, it is critical to understand how microbes in the ISS water system adapt and evolve over time. The purpose of the study is to understand the microbial evolution in the context of the complex microbial ecosystem, and to examine an influence of the spaceflight environment on microbial interactions and evolution. The hypothesis is that changes in microbial interaction between species of ISS water isolates are associated with microbial genomic evolution during spaceflight, and spaceflight is a contributing factor for the evolution of microbial interaction between different species.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	This study is searching for an understanding by which microorganisms interact and evolve in the context of the complex microbial ecosystem. While immediately applicable to the unique spaceflight environment, it will also be applicable to closed environments in which water can be recycled and reused in terrestrial environments. Overall this contributes to the creation of healthier and sustainable human built and living environments both in space and on Earth. The study uses the International Space Station as a microbial observatory to study the complex microbial ecosystems and dynamics in human built and living environments including spaceflights. The study aligns with the research goals in the Space Biology Microbiology Element to determine the effect of spaceflight on life, process, and community dynamics. Completion of this project would 1) provide genomic evidence on a potential microevolution of ISS potable water, 2) provide insight into how microorganisms in the heterogeneous population evolve over time, and 3) determine whether or not the spaceflight environment is a key contributing factor that influencing changes in microbial interaction. The knowledge gained from this study may a) fill knowledge gaps in microbial interspecies dynamics and evolution within the mixed-species population; b) provide novel insight into how the microbial ecological success within the mixed-species population contributes to microbial evolution in human Built Environment, whether in spaceflight or on Earth; and c) provide useful knowledge to support long-duration life support system, particularly, a water recovery system in spaceflight.
Task Progress:	This is the 2nd year project report. Experiment Requirements Document (ERD) baseline was completed (12/18/20), and Science Requirements Document (SRD) Approval Review was completed (01/12/21). We have tested: 1. Media composition and long-term media stability at different temperatures 2. Inoculum composition, dose, bacterial freezing media composition, inoculum preparation protocol, and frozen inoculum stability 3. Hardware chamber prototypes to determine the size, material, configuration, and design to evaluate biocompatibility and science 4. Experimental procedure including inoculation method, preflight preparation, sample collection and fixation, sample storage condition, sample process scenarios with an automation system 5. Oxygen requirement by bacteria to support optimal growth in the hardware system. The four oxygen tests will help determine the oxygen requirement to support bacterial growth and science. Tests will utilize flight-like hardware following requirements listed in the approved ERD Rev. A. We also received 16 new strains isolated from the International Space Station, from Johnson Space Center (JSC). The whole genomes of some strains are to be sequenced and analyzed. We obtained groundwater species to be used for the spaceflight experiment, from Netherlands and Sweden. The groundwater isolates are to be characterized and prepared for the spaceflight experiment. While we still have some impact from COVID-19, causing schedule delay or some challenges, our task progression is on schedule.
Bibliography Type:	Description: (Last Updated: 09/02/2025)
Articles in Peer-reviewed Journals	Yang J, Barrila J, Ott CM, King O, Bruce R, McLean RJC, Nickerson CA. "Longitudinal characterization of multispecies microbial populations recovered from spaceflight potable water." npj Biofilms and Microbiomes. 2021 Sep 6;7(1):70. https://doi.org/10.1038/s41522-021-00240-5 ; PMID: 34489467; PMCID: PMC8421509 , Sep-2021